

From Molecules to Qubits: Unravelling Noise in Self-Assembled Monolayer

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24CY40005

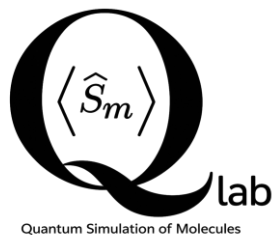
M.Sc. Chemistry, Sem-IV

Master's Thesis Project Presentation

Under the supervision of
Prof. Sabyashachi Mishra

“There's plenty of room at the bottom”
~ Richard P. Feynman, 1959

May 9, 2026



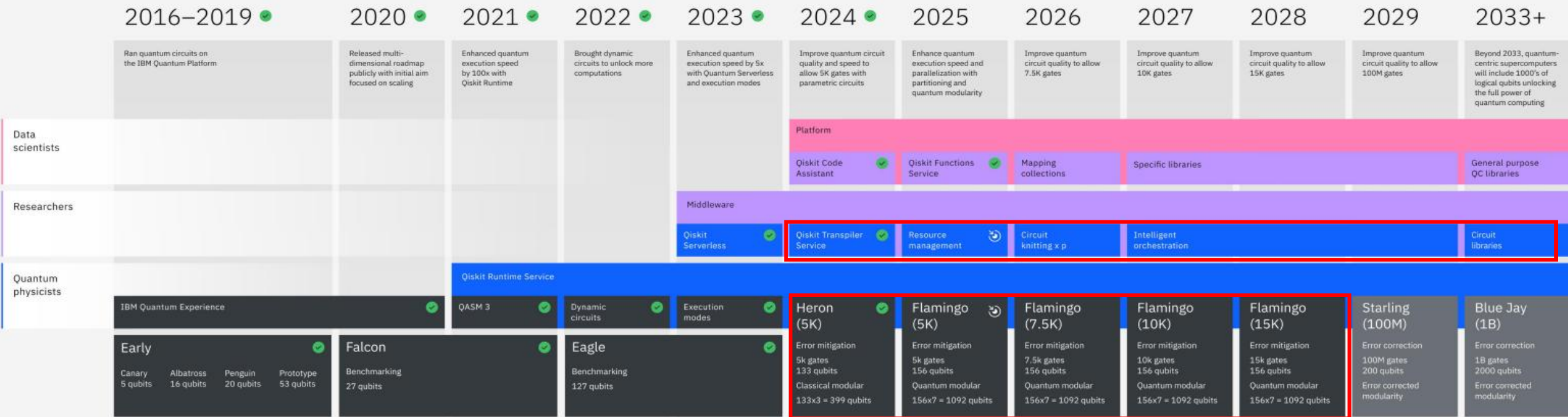
Outline

- **Introduction**
 - IBM quantum mission
 - Known facts from literature: Decoherence & Relaxation
 - Self-Assembled Monolayer (SAM)
- **Objectives**
 - Relaxation mechanism
 - Modelling TEMPO SAM
- **Computational Methods**
 - DFT methods
 - Classical simulation methods
- **Results**
 - P-TEMPO & TEMPO SAM
 - Spin-phonon coupling in P-TEMPO
 - Dynamics in TEMPO SAM
 - Spin-spin coupling in TEMPO SAM

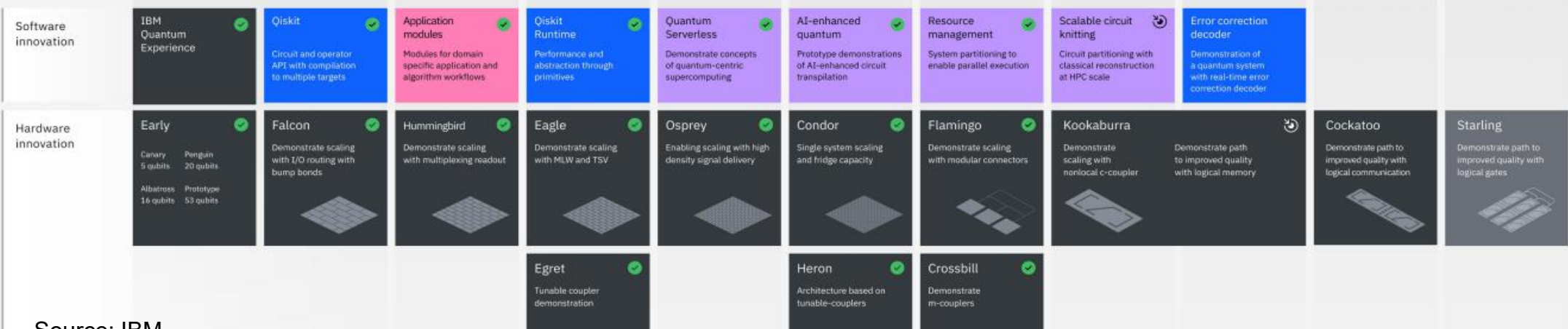
The Big Picture: IBM Quantum Mission

IBM Quantum

Development Roadmap



Innovation Roadmap



Noise !!!

Introduction: Noise

- Noise disturbs the qubit's **fragile quantum state**
- Fluctuations cause **dephasing and information loss**
- Environmental noise **destroys the phase coherence** of quantum superposition states
- Magnetic noise causes random spin, leading to **phase instability**

In the quest for quantum computers!

Device integrable qubits...

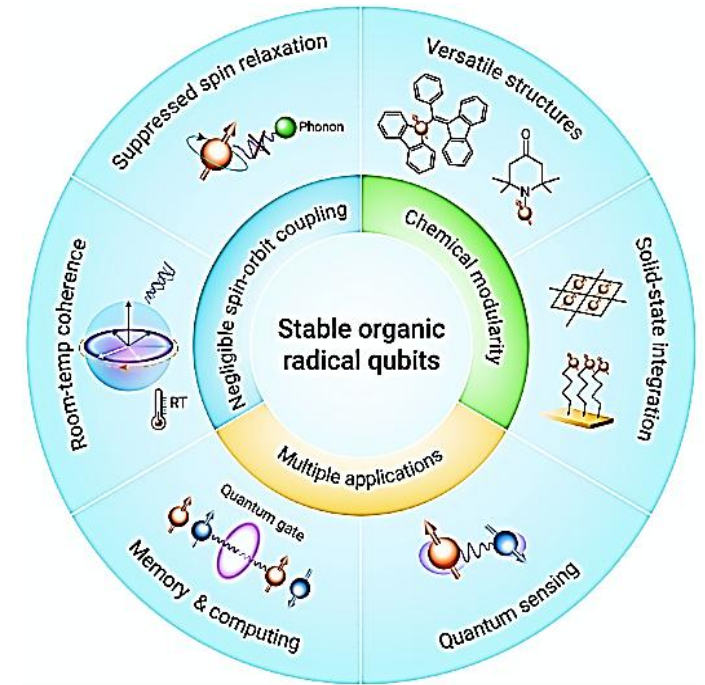
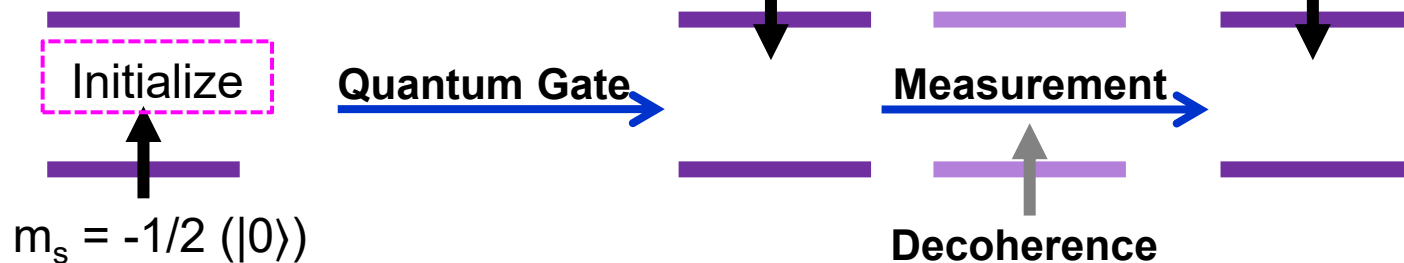
Introduction: Qubits

- Basic unit of quantum information, existing in a superposition of $|0\rangle$ and $|1\rangle$
- Scope includes faster problem-solving, secure communication and advanced sensing

DiVincenzo's criteria for quantum computation :

1. A **scalable** physical system with well-characterized qubit
2. The ability to **initialize** the state of the qubits to a simple fiducial state
3. Long relevant quantum **coherence** times
4. A universal set of quantum **gates**
5. A qubit-specific **measurement** capability (readout)

$$m_s = +1/2 (|1\rangle)$$

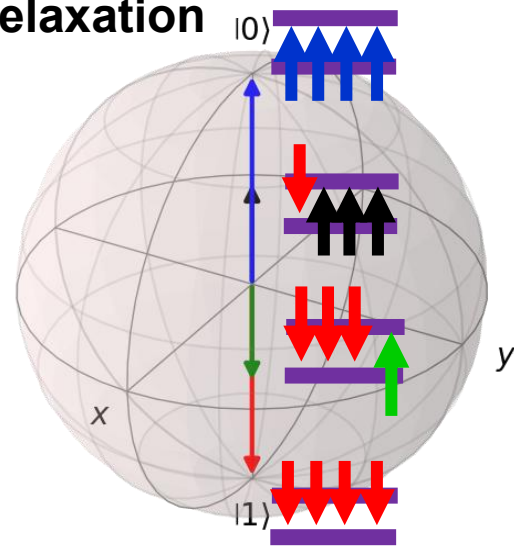


Stable Organic Radical Qubits and their Applications in Quantum Information Science

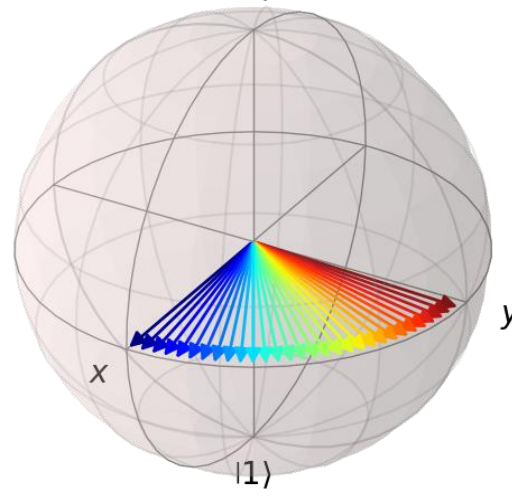
A. Zhou, Z. Sun, and L. Sun, *The Innovation*, **2024**, 5, 5, 100662

Introduction: Relaxation Mechanism

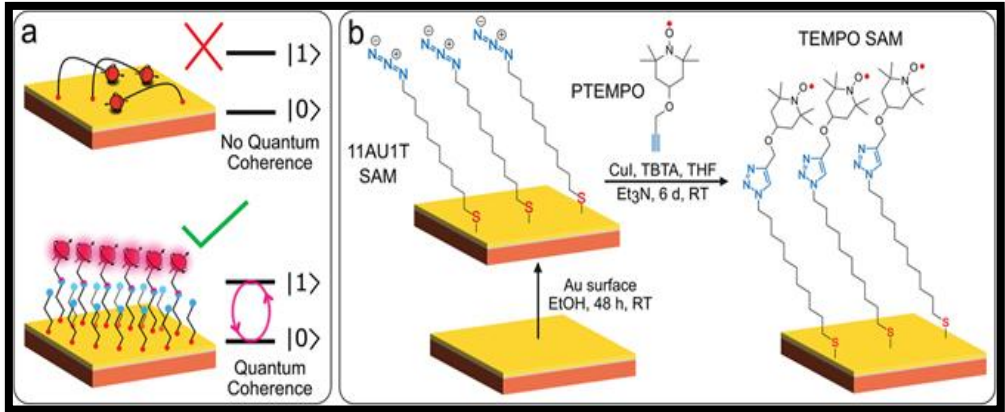
Spin relaxation



Decoherence



TEMPO SAM



Process	Reasons
Spin Relaxation	1. Spin-Spin Interaction 2. Spin-Lattice Interaction
Spin Decoherence	Environmental Magnetic Noise

System	$T_m(\mu s)$
TEMPO SAM	13.53
P-TEMPO (dilute solution)	3.23

L. Tesi, F. Stemmler, M. Winkler, S. S. Y. Liu, S. Das, X. Sun, M. Zharnikov, S. Ludwigs, and J. V. Slagereen, *Advanced Materials*, **2023**, 35, 10, 2208998

Objectives

- **Analyze vibrational effects**, especially N–O bond distortions, on the **g-tensor** and **A-tensor**
- **Identify noise sources** in SAM qubits: spin-spin interactions
- **Spin dynamics of radicals in SAM**: how does the local environment affect coupling
- **Correlate molecular arrangement**: decoherence pathways and qubit stability

Finding spin-phonon coupling contribution of vibrational modes of P-TEMPO

Modelling TEMPO SAM using molecular simulation methods

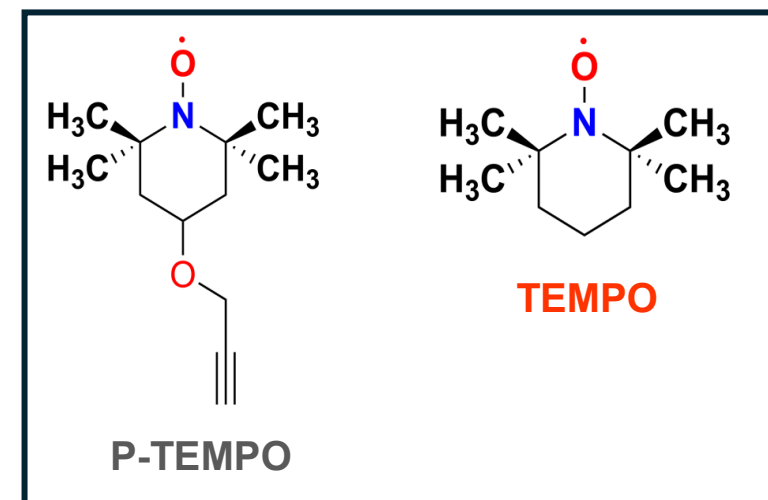
Analyzing the difference in spin-dynamics of PTEMPO and TEMPO SAM

Computational Methodology: P-TEMPO

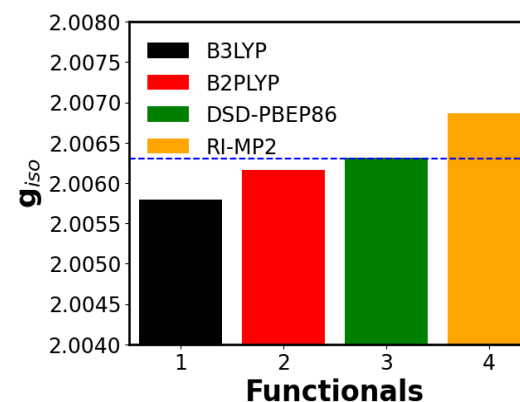
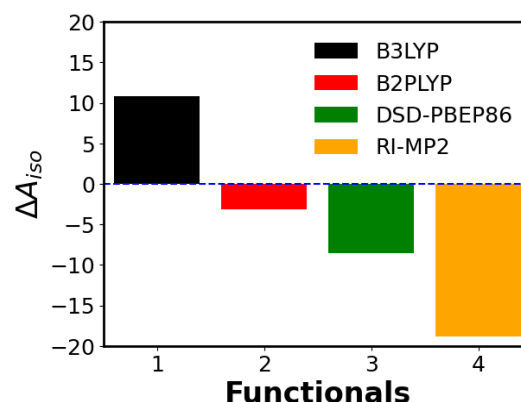
Geometry
Optimization



Theory:
Functional: uB3LYP
Basis set: def2-TZVP



Spin-Phonon
Coupling



Functional Benchmarking:

- B3LYP
- **B2PLYP**
- DSD-PBEP86
- RI-MP2

Basis set: EPR-II



Computational Methodology: SAM

Classical Molecular Dynamics Simulation :

- Self-assembled monolayer with P-TEMPO radicals **has more than 12500 atoms**
- Quantum mechanical simulations are **very costly**

Initial coordinates:

in-house Python script

- Build an Au (111) surface using VESTA
- Preparation of alkanethiolate TEMPO
- Experimental surface coverage:
 - 4×10^{14} molecules/cm²
Tesi et. al., *Adv. Mat.*, 2023
 - **141 molecules in 3525 Å² (model)**
- All TEMPO molecules: placed normally

- Interfacial force field for Au
- General Amber Force Field for C, H, N, O, S

Minimization: 10000 steps



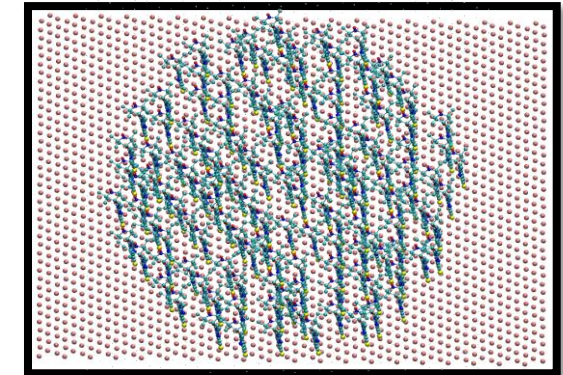
Heating: 0-300 K, 1 ns



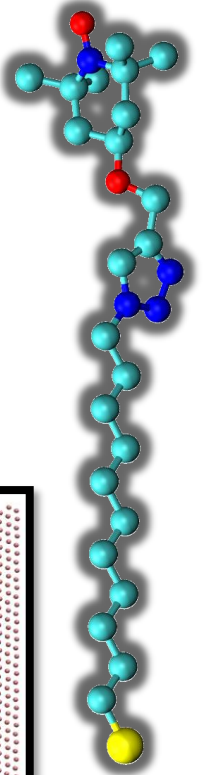
Production: 10 ns, 300 K

Temperature:

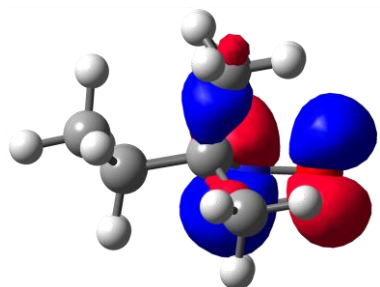
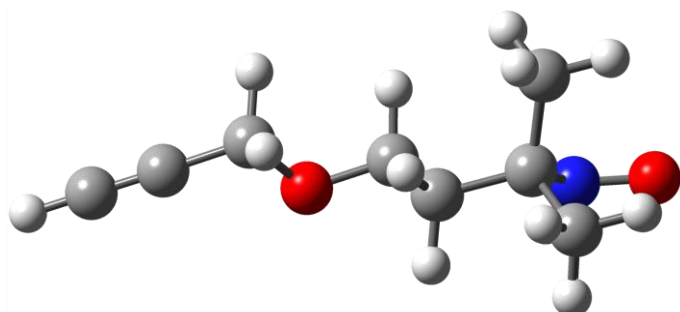
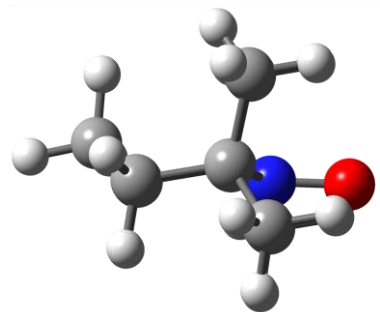
- 300 K
- 100 K



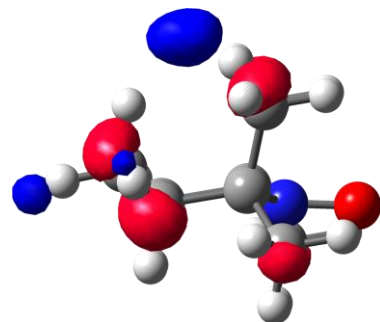
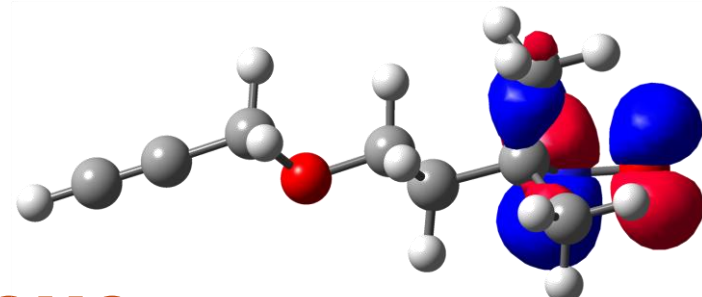
- Langevin thermostat
- NVT ensemble
- Vacuum
- Au and S atoms restrained



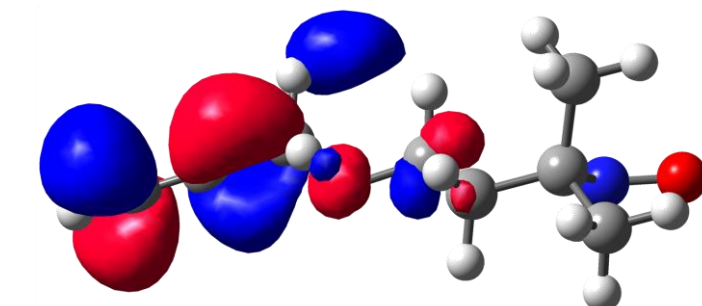
Results: TEMPO & P-TEMPO - Electronic Structure



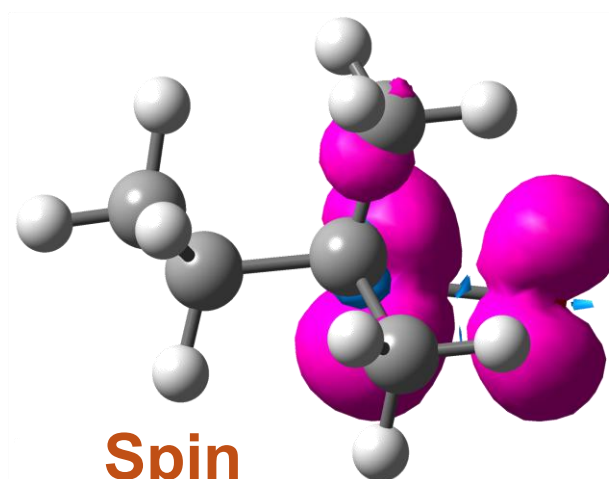
HOMO



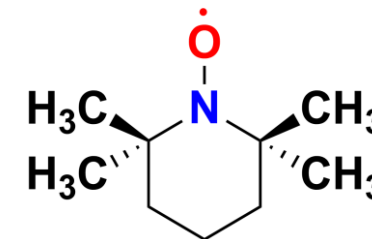
LUMO



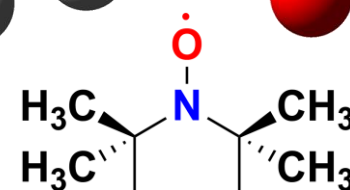
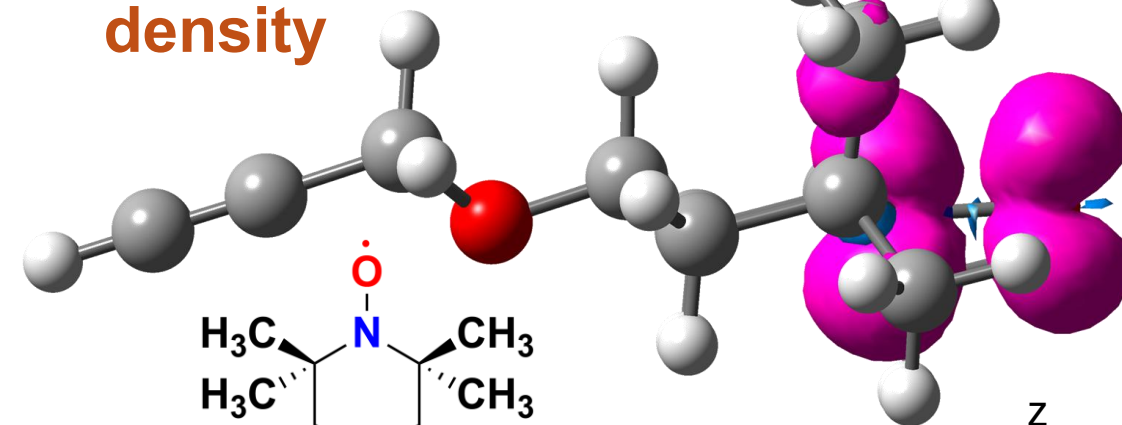
uB3LYP/ def2-TZVP



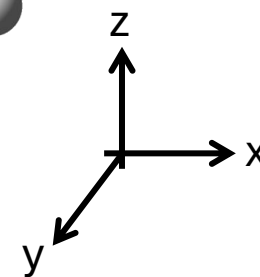
Spin
density



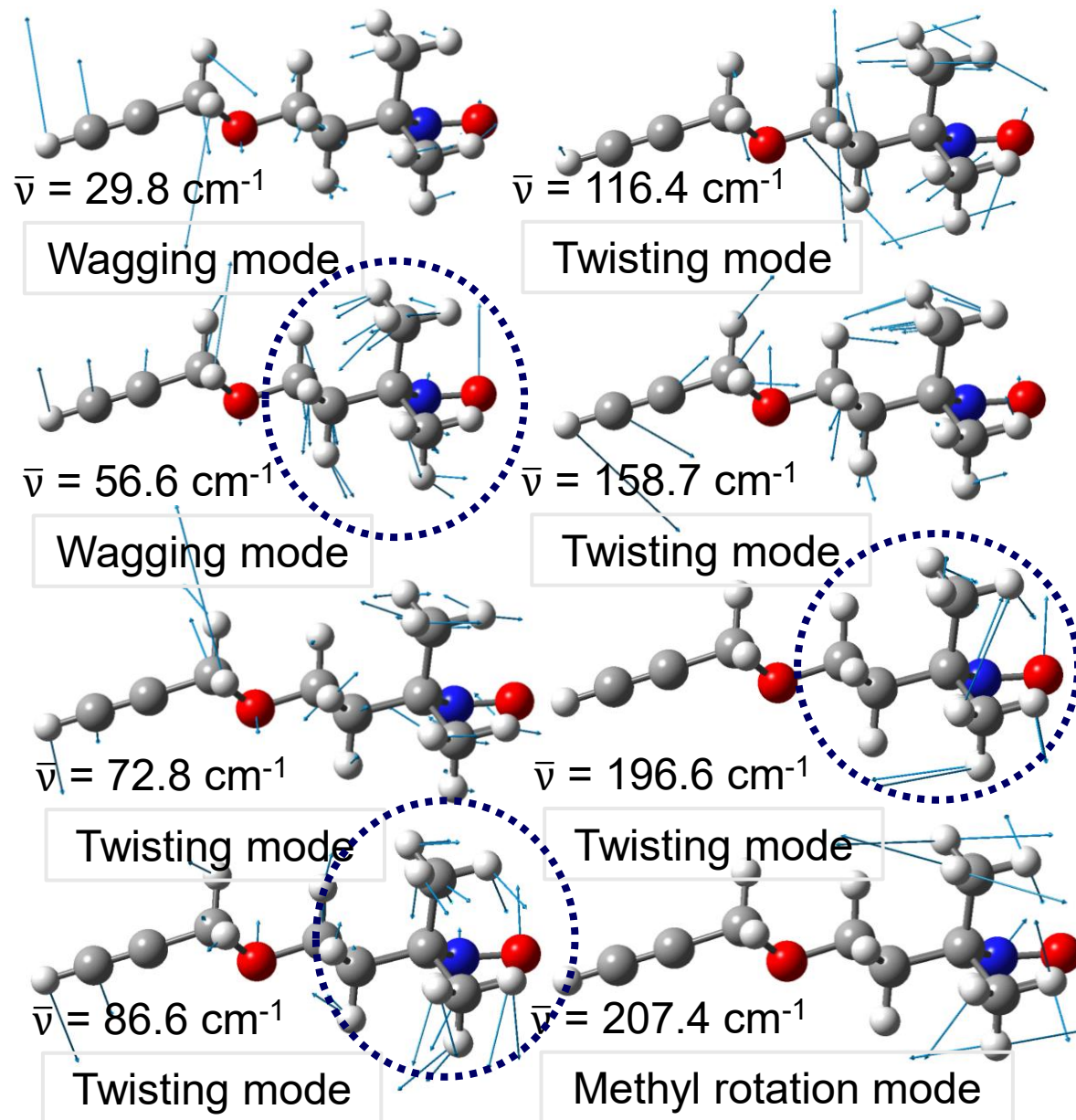
TEMPO



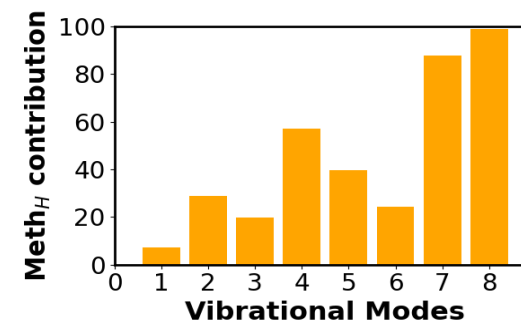
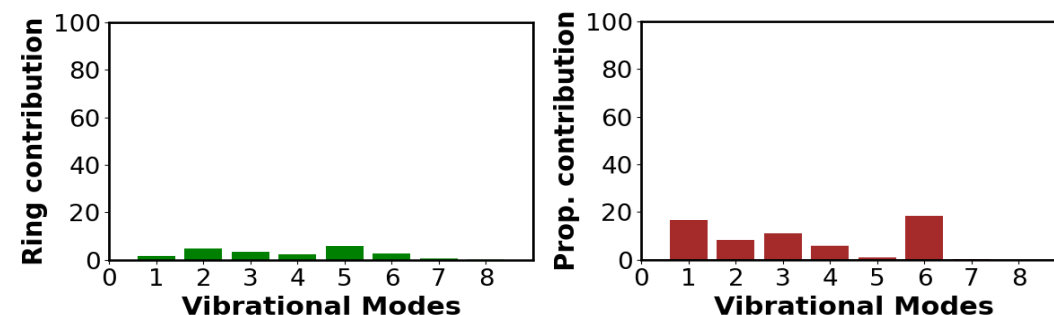
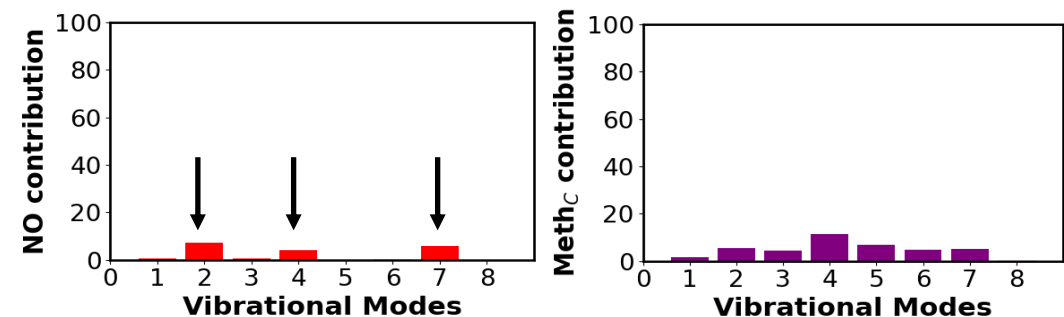
P-TEMPO



Results: Vibrational Modes in P-TEMPO



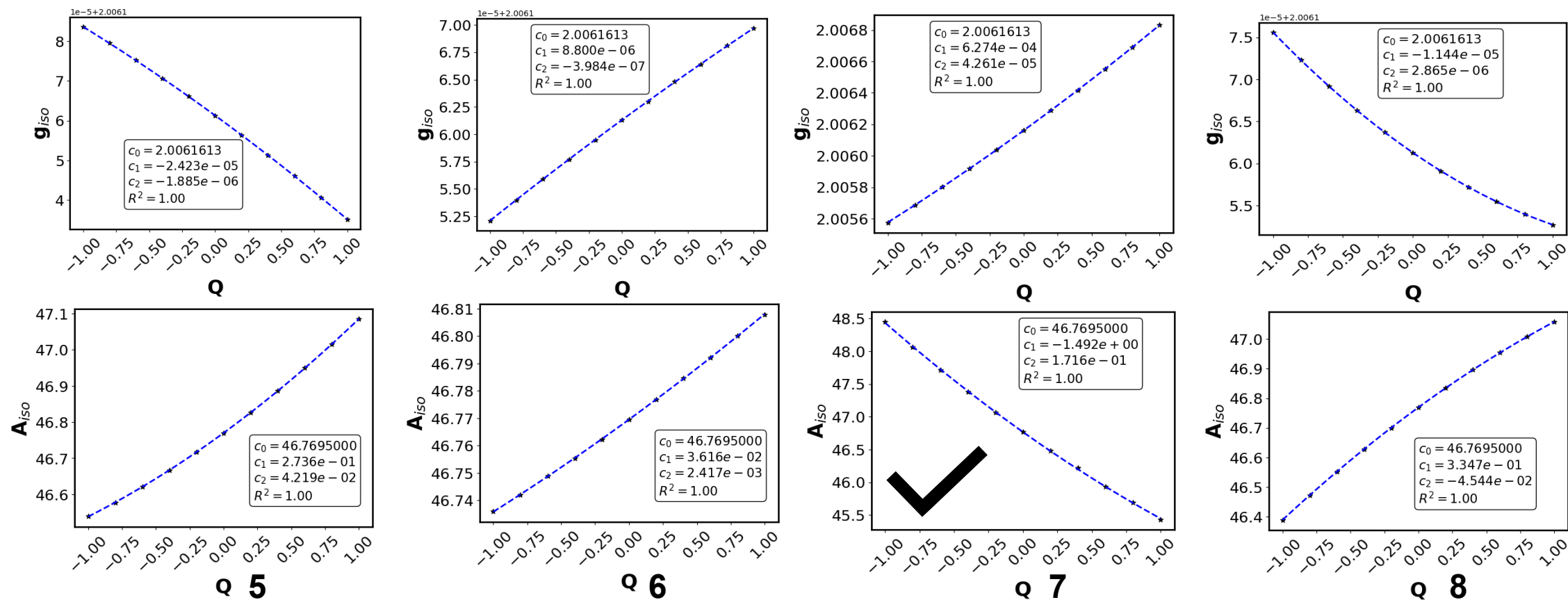
- Thermal energy available at r.t. (300 K) = $k_B T$
- Approx. freq. corresponding to 300 K is 208.5 cm^{-1}



Results: g & A Anisotropies in P-TEMPO

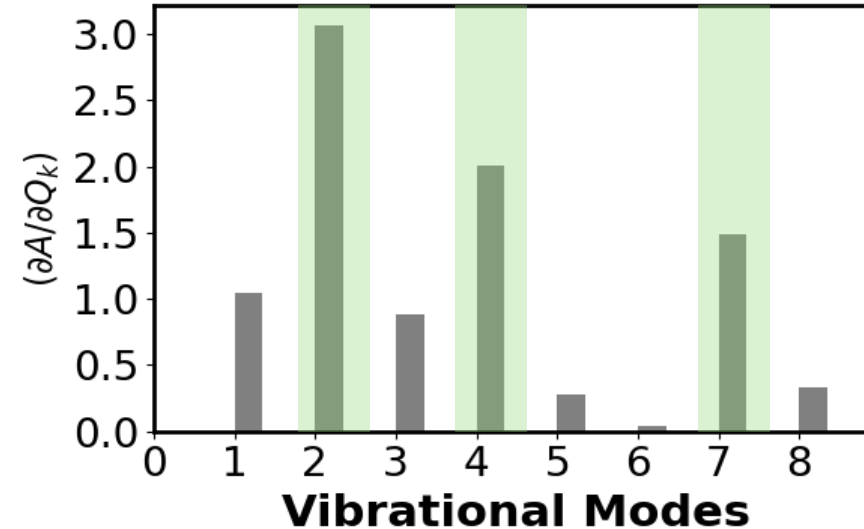
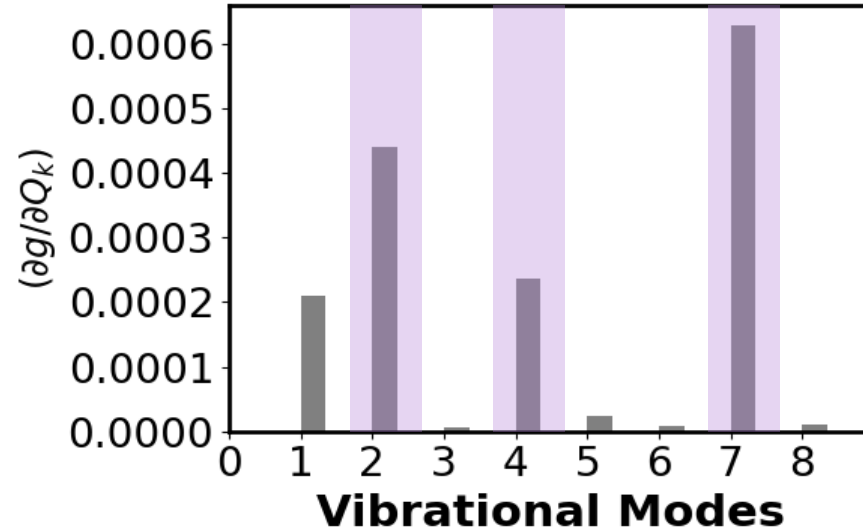
$P = g \text{ or } A$

$$P(\{Q_k\}) = P^{(0)} + \sum_k \left(\frac{\partial P}{\partial Q_k} \right)_0 Q_k + \frac{1}{2} \sum_{k,l} \left(\frac{\partial^2 P}{\partial Q_k \partial Q_l} \right)_0 Q_k Q_l + \dots$$



- Quadratic dependence of g on Q (Dominant SOC)
- Linear dependence of g on Q (Dominant environmental perturbation)

Results: Spin-Phonon Coupling in P-TEMPO



Modes 2, 4 & 7 have significant contribution from N—O vibrations

- **Strong mode selectivity:** Only a few vibrational modes contribute significantly
- **Dominant relaxation channel:** Mode 2 shows the highest coupling (**Expt. verified-** Mode around 58 cm⁻¹)
- **Local vibrations matter:** Mainly modulated by specific molecular motions
- **Decoherence pathway:** Spin relaxation governed by limited phonon modes

Results: Modelling T_1 relaxation in P-TEMPO

$$\frac{1}{T_1} = A \frac{\exp\left(\frac{\hbar\omega_1}{k_B T}\right)}{\exp\left(\frac{\hbar\omega_1}{k_B T}\right) - 1} + B \sum_i V_i \frac{\exp\left(\frac{\hbar\omega_i}{k_B T}\right)}{\left(\exp\left(\frac{\hbar\omega_i}{k_B T}\right) - 1\right)^2} + C$$

Field-dependent
(Direct)

Multi-mode phonon-driven
(Spin-vibrational coupling)

T
independent

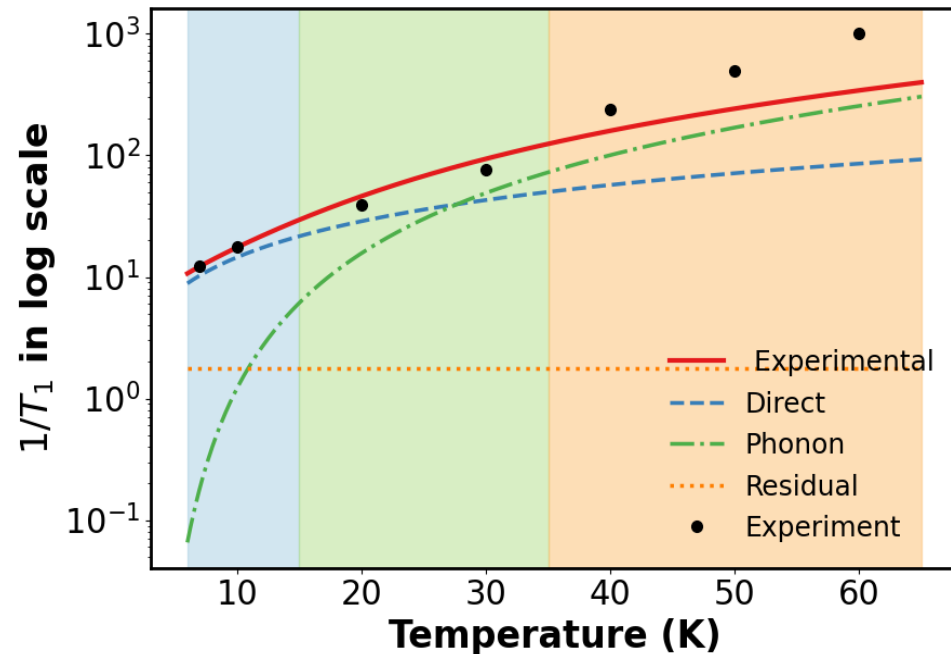
T_1 : Spin-lattice relaxation time

A, B, & C : Strengths of the direct, phonon-mediated, and residual relaxation processes

ω_1 : Zeeman frequency

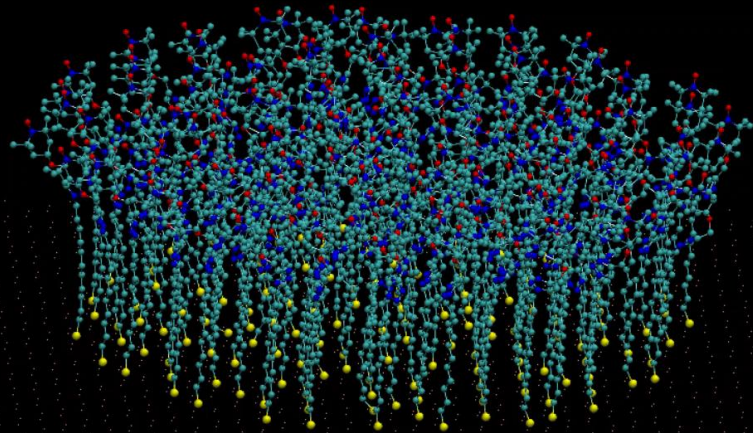
ω_i : Normal mode frequencies

V_i : Spin-phonon coupling constant

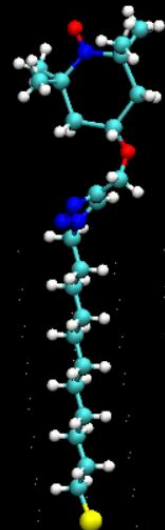
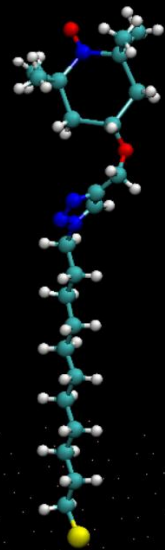


Temperature Range	Direct (%)	Multi-mode Phonon (%)	Residual (%)
Low (7-15 K)	79.7	11.5	8.8
Medium (15-35 K)	50.6	46.9	2.4
High (35-65 K)	28.7	70.6	0.7

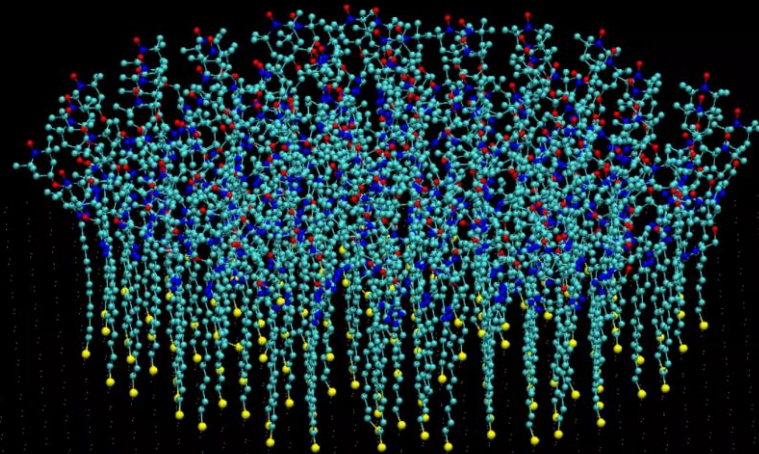
Results: Simulation of TEMPO SAM



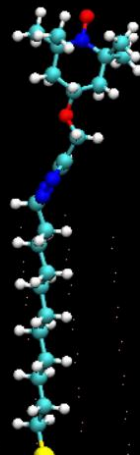
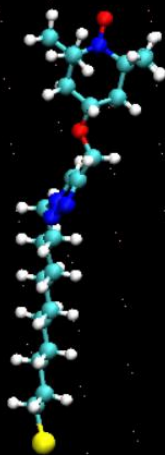
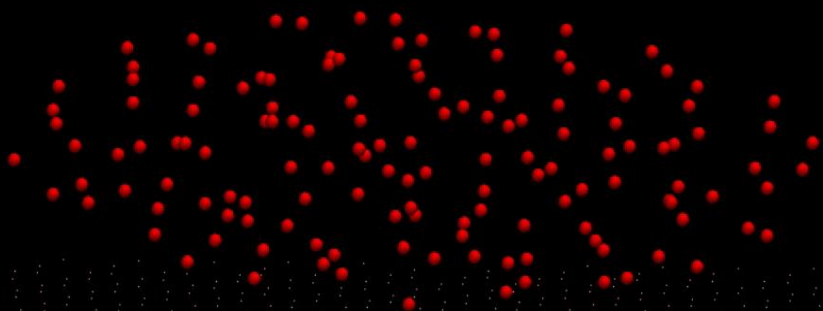
300 K



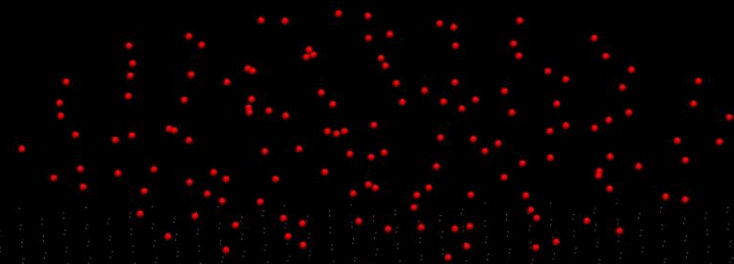
Central TEMPO



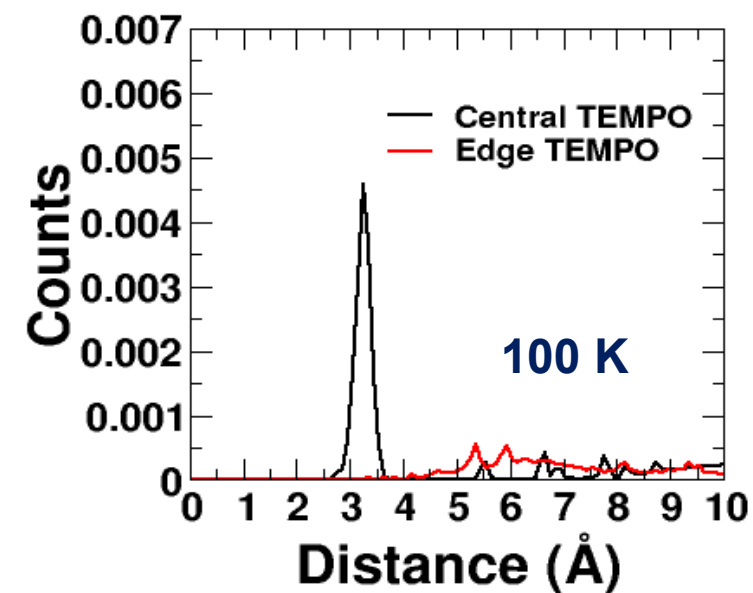
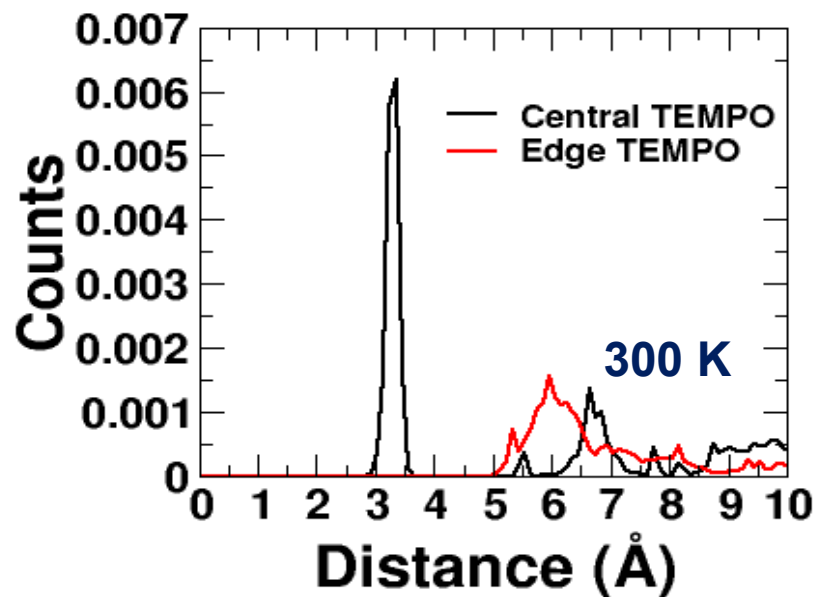
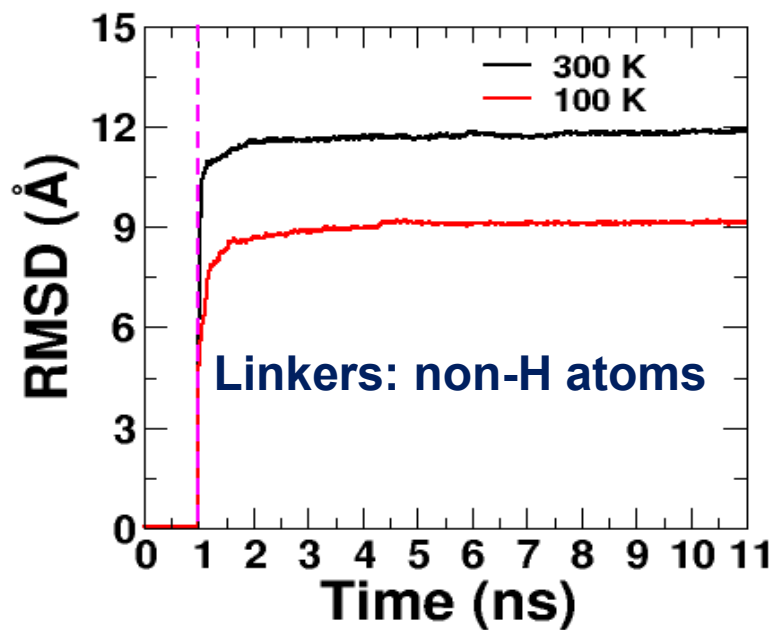
100 K



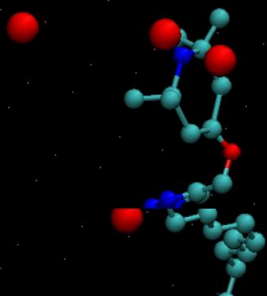
Edge TEMPO



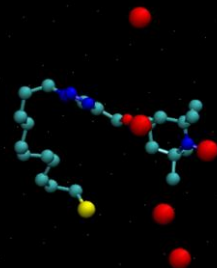
Results: Spin Dynamics in TEMPO SAM



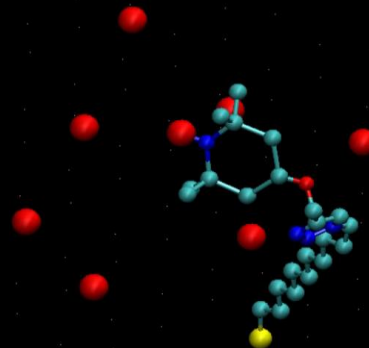
300 K
Central TEMPO



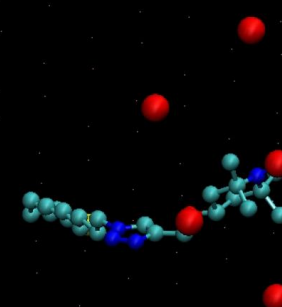
300 K
Edge TEMPO



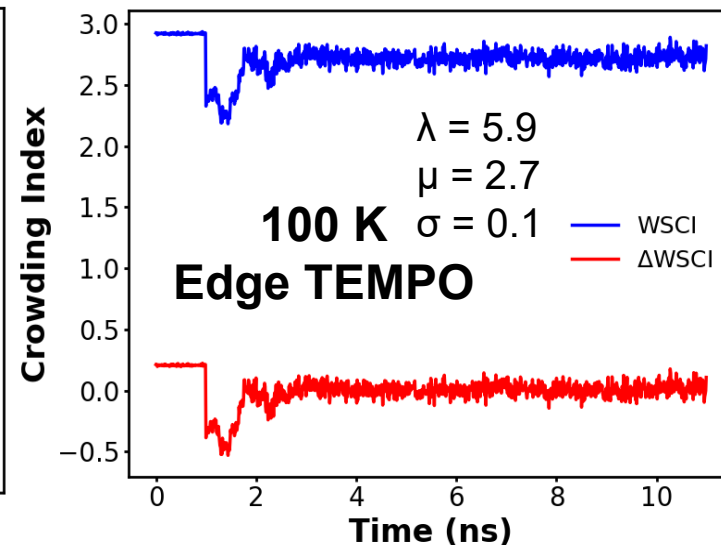
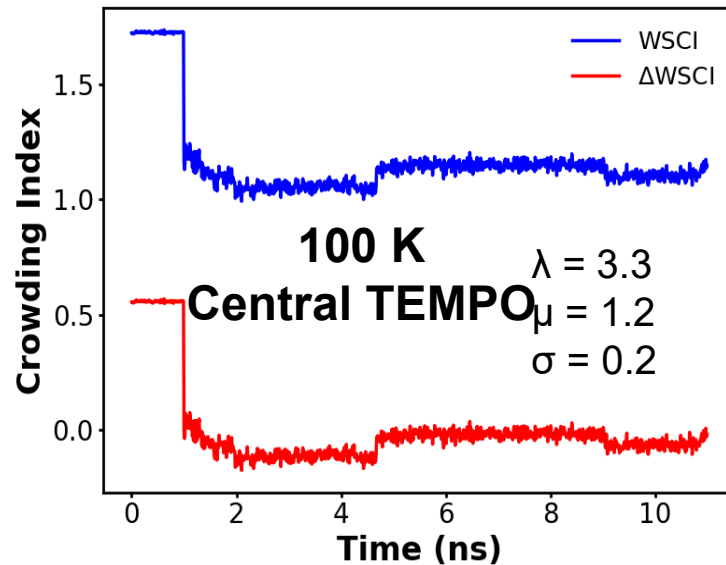
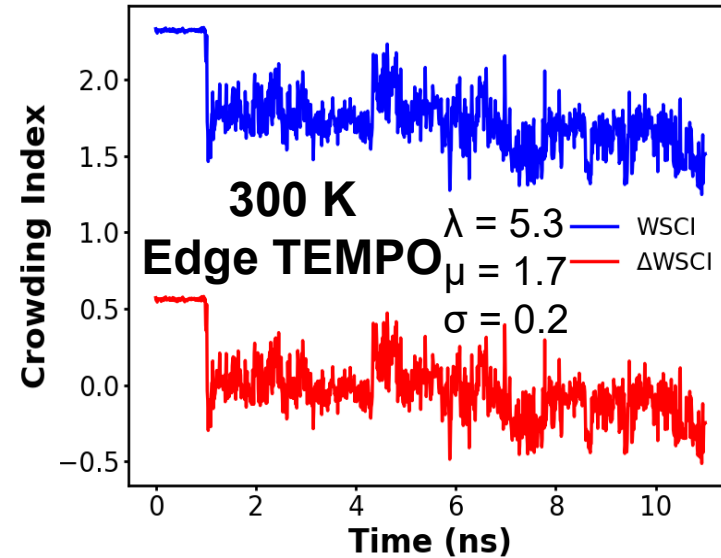
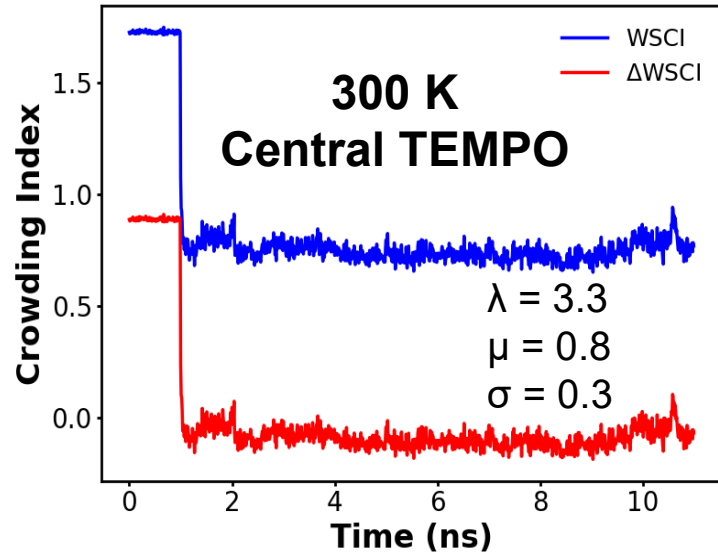
100 K
Central TEMPO



100 K
Edge TEMPO



Results: Spin Dynamics in TEMPO SAM



Weighted Spin Crowding Index (WSCI)

$$\eta_{CI}^{(i)}(t) = \sum_{j \neq i} e^{-r_{ij}(t)/\lambda}$$

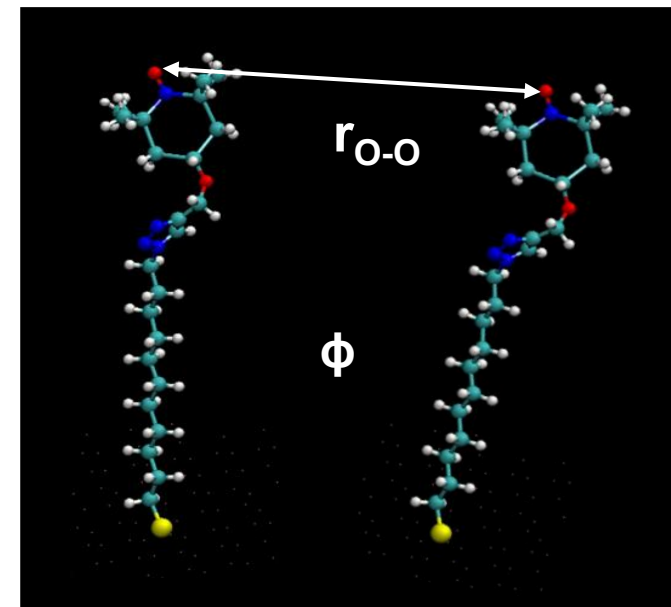
$$\langle \eta_{CI} \rangle = \frac{1}{T} \sum_{t=1}^T \eta_{CI}^{(i)}(t)$$

$$\Delta WSCI_i(t) = \eta_{CI}^{(i)}(t) - \langle \eta_{CI} \rangle$$

Feature	Central TEMPO	Edge TEMPO
Crowding	High	Low
Fluctuations	Small	Large
Stability	High	Moderate
Noise level	Low	High
Coherence	Long	Short

Results: Spin-Spin Coupling in TEMPO SAM

- **Minimum energy structure**
- Select **one central TEMPO radical**
- Compute interactions with **neighbouring radicals**
- Measure:
 - O–O distance** (intermolecular separation)
 - O–N–N–O dihedral angle** (relative orientation)
- Perform broken symmetry calculations



$$J_{AB} = -\frac{E_{HS} - E_{BS}}{\langle S^2 \rangle_{HS} - \langle S^2 \rangle_{BS}}$$

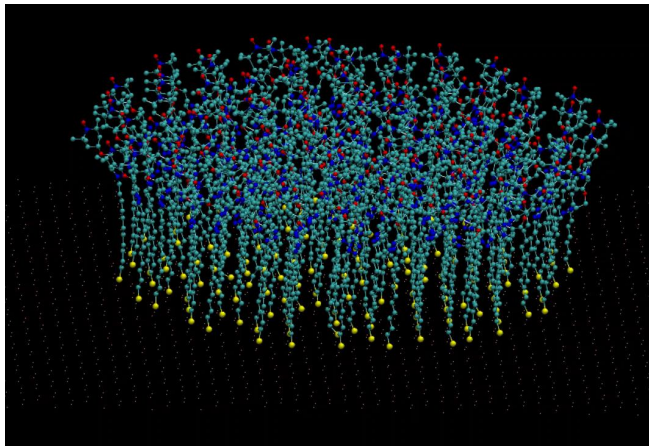
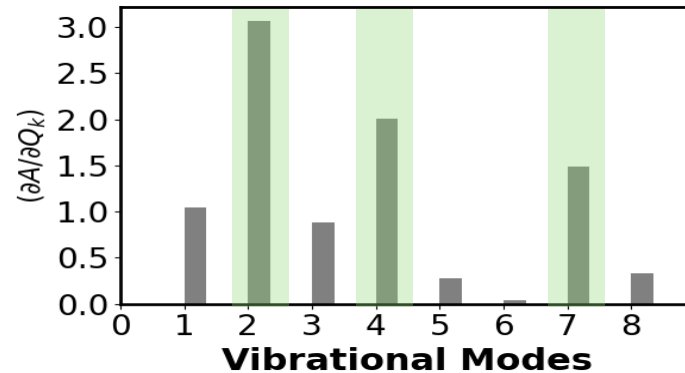
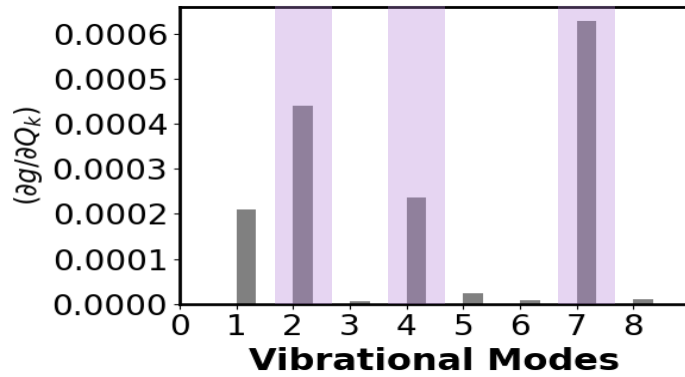
Distance (Å)	Angle(°)	J (cm ⁻¹)
3.56	-22.45	-1263.47
4.84	105.13	-0.23
5.66	5.64	0.01

Central spin centres are **shielded** by the neighbouring shell via antiferromagnetic coupling

Looking forward...

- **Static geometry approximation:** Minimum energy structure
- **Simplified environment:** Long-range interactions treated classically, not QM derived
- **Limited description of surface effects:** Surface-induced fluctuations not considered

Wrapping up



- Only a **few vibrational modes** contribute to spin-phonon coupling
- Dynamics of **central vs edge** radicals in SAM
- J **decreases** sharply with increasing O–O distance
- The O–N–N–O **dihedral** strongly affects the **sign** and **magnitude** of J

Feature	Central TEMPO	Edge TEMPO
Mean crowding	High	Low
Fluctuations	Small	Large
Stability	High	Moderate
Noise level	Low	High
Coherence	Long	Short

*S.Karmakar, **A.A.Begg**, M.Nath, and S.Mishra, 2026 (Manuscript ready for submission)*

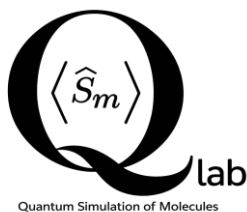
Acknowledgements

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- **Head, Dept. of Chemistry:** Prof. Sanjoy Bandyopadhyay
- **Fac. Adv.:** Dr. Pradip Kumar Chakraborty
- Prof. Lorenzo Tesi, University of Stuttgart, Germany
- **Special thanks:** Ms. Moromi Nath
- QSM lab members
- Friends & Family



QSM Lab, Dept. of Chemistry, IIT Kharagpur
Image courtesy: M.Nath

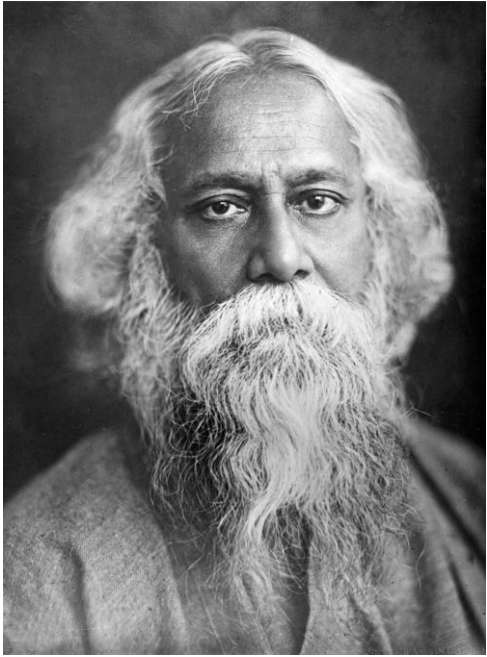
Paramshakti Supercomputing Facility
National Supercomputing Mission



Special thanks to the panel members

Thank you for your kind attention!





Courtesy: Wikipedia

"The real purpose of education is not to fill your head with facts, but to light a fire of curiosity"

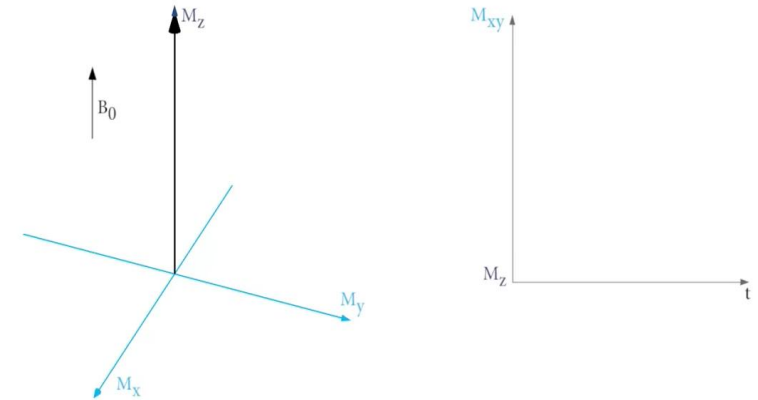
~Rabindranath Thakur

Appendix

A 1: Relaxation Times

T_1 (Spin-Lattice Relaxation): How fast the spin system exchanges energy with the lattice (along z-axis)

T_2 (Spin-Spin Relaxation): How fast spin coherence is lost due to random interactions between spins. Measures the decay of transverse magnetisation (in the xy-plane)



Courtesy: Wikipedia

T_1 : how fast the spins return to equilibrium?

T_2 : how fast the spins lose phase sync?

A 2: Broken Symmetry Calculations

- Based on **spin-unrestricted wavefunction** where spins on different centres are **intentionally flipped** ($\uparrow\downarrow$)
- BS state is **not a pure spin state**, but approximates the mixed-spin situation
- Exchange coupling J_{AB} is obtained by comparing **high-spin (HS)** and **BS energies**

$$\hat{H} = -2J_{AB} \vec{S}_A \cdot \vec{S}_B$$

$$S = S_A + S_B \text{ to } |S_A - S_B|$$

$$J_{AB} = - \frac{E_{HS} - E_{BS}}{\langle S^2 \rangle_{HS} - \langle S^2 \rangle_{BS}}$$

$J_{AB} > 0$: ferromagnetic coupling; highest spin state is lowest in energy

$J_{AB} < 0$: antiferromagnetic coupling; lowest spin state is lowest in energy.